

Jarvis Voice Assistant using Python

Project Overview

Jarvis is a Python-based voice assistant that can perform various tasks through voice commands or text commands. The assistant uses speech recognition, text-to-speech technology, and web automation to provide an interactive user experience.

Unlike traditional implementations that depend on PyAudio, this project uses **Sounddevice** for recording audio, making it compatible with newer Python versions such as Python 3.14.

Features

Voice Recognition

- Listen to user commands using a microphone.
- Convert speech into text using Google Speech Recognition.
- Support for text mode if the microphone is unavailable.

Text-to-Speech

- Speak responses using `pyttsx3`.
- Read jokes, Wikipedia summaries, and user-provided text.

Personal Information

- Remember the user's name.
- Recall the user's name later.

Date and Time

- Tell the current time.
- Tell the current date.

Web Automation

- Open YouTube.
- Open Google.
- Open GitHub.
- Open Stack Overflow.

Application Automation

- Open Notepad.

- Open Calculator.

Search and Information

- Search Google.
- Search Wikipedia.

Entertainment

- Play songs on YouTube.
- Tell English jokes.
- Tell Hindi jokes.

Text Reading

- Read any text spoken by the user.
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Technologies Used

- Python 3.14
 - Sounddevice
 - SpeechRecognition
 - Pyttsx3
 - Wikipedia
 - PyWhatKit
 - PyJokes
 - Scipy
 - JSON
-

Project Structure

```
Jarvis/  
|  
├─ jarvis.py  
├─ README.md  
├─ requirements.txt  
└─ .jarvis_data.json
```

Installation

Step 1: Clone the Repository

```
git clone https://github.com/your-username/jarvis-assistant.git  
cd jarvis-assistant
```

Step 2: Create Virtual Environment

Windows

```
python -m venv venv  
venv\Scripts\activate
```

Linux

```
python3 -m venv venv  
source venv/bin/activate
```

Step 3: Install Required Packages

```
pip install sounddevice  
pip install scipy  
pip install SpeechRecognition  
pip install pyttsx3  
pip install wikipedia  
pip install pywhatkit  
pip install pyjokes
```

Or:

```
pip install -r requirements.txt
```

requirements.txt

```
sounddevice  
scipy  
SpeechRecognition  
pyttsx3  
wikipedia  
pywhatkit  
pyjokes  
numpy
```

Running the Project

Voice Mode

```
python jarvis.py
```

Text Mode

```
python jarvis.py --text
```

Disable Speech Output

```
python jarvis.py --no-speak
```

Supported Commands

Greeting Commands

```
hello  
hi  
hey
```

User Information

my name is Siddhnath
what is my name

Date and Time

what is the time
tell me the date

Open Websites

open youtube
open google
open github
open stackoverflow

Open Applications

open notepad
open calculator

Search Commands

search for python programming

Wikipedia

who is Narendra Modi
what is artificial intelligence
wikipedia python

Music

play shape of you
play beliver song

Jokes

tell me a joke
say a joke
hindi joke
another joke

Read Text

say hello everyone
speak welcome to my project
read this sentence

Exit

exit
quit
stop
goodbye

How It Works

Voice Input

1. The microphone records audio using Sounddevice.
2. Audio is temporarily stored in WAV format.
3. SpeechRecognition converts the audio into text.

Command Processing

1. User input is converted to lowercase.
2. Regular expressions identify commands.
3. Corresponding actions are executed.

Text-to-Speech

The assistant responds using pyttsx3.

Modules Description

load_data()

Loads saved user information.

save_data()

Stores user information in JSON format.

say()

Prints and speaks text.

hear_once()

Records and recognizes voice commands.

handle_command()

Processes all user commands.

run()

Runs the assistant in text mode or voice mode.

main()

Parses command-line arguments and starts Jarvis.

Future Enhancements

- Weather Forecast API
 - Email Sending
 - WhatsApp Messaging
 - Face Recognition
 - Object Detection
 - ChatGPT Integration
 - System Control Commands
 - Reminder System
 - Alarm Clock
 - GUI Interface
 - Database Integration
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Advantages

- Easy to use.
 - Works on Python 3.14.
 - Does not require PyAudio.
 - Supports voice and text mode.
 - Cross-platform compatibility.
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Limitations

- Requires internet for Google Speech Recognition.
 - Limited command set.
 - Fixed recording duration.
 - No wake-word detection.
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Author

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License

This project is developed for educational and learning purposes.